

9. (a) Ethanedioic acid (oxalic acid) has the formula $(\text{COOH})_2$.

It can be made in the laboratory by oxidising sugar with a powerful oxidising agent (shown as $[\text{O}]$ in the equation below).

- (i) Balance the equation. [1]



- (ii) After purification, the oxalic acid is produced as white crystals of a hydrate, $(\text{COOH})_2 \cdot x\text{H}_2\text{O}$. On heating this hydrate, water is lost, leaving anhydrous oxalic acid.



In an experiment 5.67 g of the hydrate were heated to constant mass, giving 4.05 g of the anhydrous acid.

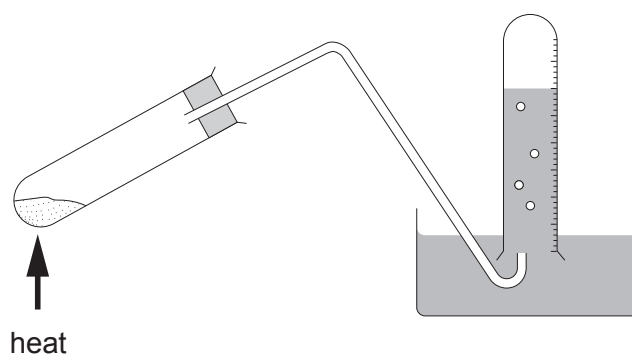
Calculate the value of x in $(\text{COOH})_2 \cdot x\text{H}_2\text{O}$. You **must** show your working. [3]

$x = \dots\dots\dots$

- (b) Another method of producing oxalic acid in the laboratory is to heat potassium methanoate, HCOOK . This produces potassium oxalate, which is then acidified to give the acid.



Some students carried out this experiment using the apparatus below. They collected the hydrogen gas produced over water.



A few students obtained a smaller volume of hydrogen than expected.

Suggest **two** practical reasons why this may have occurred, apart from leaks. [2]

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- (c) Gareth was asked to analyse a mixture of potassium methanoate and potassium oxalate. He weighed out 4.69 g of the mixture and carried out the method below.

- The mixture was completely dissolved in distilled water.
- Calcium chloride solution was added to this solution (only calcium oxalate was precipitated).
- The mixture was filtered and the precipitated calcium oxalate washed and dried.

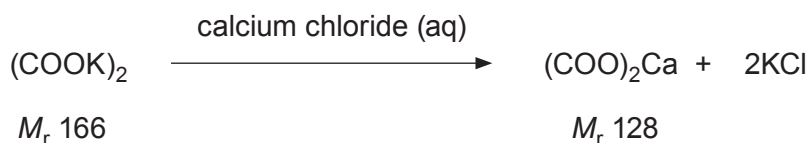
- (i) Suggest how Gareth would know when enough calcium chloride solution had been added to react with all of the potassium oxalate present. [2]

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- (ii) In his experiment Gareth obtained 2.49 g of pure dry calcium oxalate (M_r 128).



Calculate the percentage of potassium oxalate in the mixture, giving your answer to the appropriate number of significant figures. [3]

Percentage of potassium oxalate = %



9. (a) A student is given five beakers labelled **A**, **B**, **C**, **D** and **E**. Each contains a different solution. The solutions are barium bromide, barium nitrate, calcium chloride, magnesium chloride and magnesium nitrate but it is not known which beaker contains which solution.

She is also given solutions of three reagents: silver nitrate, sodium hydroxide and sulfuric acid.

Devise a plan that uses the reagents to unambiguously determine which solution is in which beaker.

You should include the observations that enable you to positively identify **all** five solutions.
[6 QER]

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- (b) Give a test which would confirm the presence of calcium ions in aqueous calcium chloride. State the result of the test. [2]

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- (c) A solution is known to contain a mixture of sodium carbonate and sodium nitrate only. To determine the carbonate ion concentration in the mixture, aqueous barium chloride was added in excess. The precipitate produced was filtered off and dried by strong heating.

- (i) Write the **ionic** equation, including state symbols, for this precipitation reaction. [1]

- (ii) State why the aqueous barium chloride was added in excess. [1]

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- (d) All Group 2 carbonates are insoluble and can be precipitated, dried and weighed in experiments similar to that in part (c). However, the strong heating needed to drive off all the water can cause a problem in accurately determining the mass of the carbonate precipitated. Suggest what this problem might be. [2]

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(e) Explain the following observations by reference to the bonding present in each of the substances.

(i) Ionic substances such as calcium chloride can conduct electricity under certain circumstances. [2]

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(ii) Iodine is a solid which vaporises on gentle warming. [2]

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(iii) A metal such as magnesium is malleable. [2]

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END OF PAPER

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Surname	Centre Number	Candidate Number
Other Names		2

**GCE AS/A LEVEL**

2410U10-1



S18-2410U10-1

CHEMISTRY – AS unit 1**The Language of Chemistry, Structure of Matter and Simple Reactions**

TUESDAY, 22 MAY 2018 – MORNING

1 hour 30 minutes

Section A**Section B**

For Examiner's use only		
Question	Maximum Mark	Mark Awarded
1. to 6.	10	
7.	12	
8.	15	
9.	15	
10.	13	
11.	15	
Total	80	

2410U101
01**ADDITIONAL MATERIALS**

In addition to this examination paper, you will need a:

- calculator;
- **Data Booklet** supplied by WJEC.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen or correction fluid.

Write your name, centre number and candidate number in the spaces at the top of this page.

Section A Answer **all** questions in the spaces provided.**Section B** Answer **all** questions in the spaces provided.Candidates are advised to allocate their time appropriately between **Section A (10 marks)** and **Section B (70 marks)**.**INFORMATION FOR CANDIDATES**

The number of marks is given in brackets at the end of each question or part-question.

The maximum mark for this paper is 80.

Your answers must be relevant and must make full use of the information given to be awarded full marks for a question.

The assessment of the quality of extended response (QER) will take place in **Q.11(a)(i)**.

If you run out of space, use the additional page(s) at the back of the booklet, taking care to number the question(s) correctly.



MAY182410U10101

9. Ascorbic acid, $C_6H_8O_6$, is the main component of vitamin C tablets. Its name is derived from *a-* (meaning “no”) and *scorbutus* (scurvy), the disease caused by a deficiency of vitamin C. A student was asked to find the percentage of ascorbic acid in identical vitamin C tablets.

She was told to use the following method.

- Fill a burette with $0.100 \text{ mol dm}^{-3}$ sodium hydroxide solution.
- Weigh a conical flask and record its mass.
- Add a vitamin C tablet to the flask, reweigh it and record its mass.
- Add about 50 cm^3 of deionised water to the flask and swirl to break up the tablet.
- Heat the flask gently for 5 to 10 minutes.
- After the solution has cooled add a few drops of a suitable indicator.
- Carry out a rough titration of this solution with the sodium hydroxide solution.
- Accurately repeat the procedure several times and calculate a mean titre.

- (a) A **three** decimal place balance was used. The mass of each vitamin C tablet was 500 mg.

Calculate the maximum percentage error in the weighing of the tablet.
You **must** show your working.

[2]

Maximum percentage error = %

- (b) (i) Suggest why she did not need to measure the volume of water accurately. [1]

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- (ii) Suggest why she heated the flask for 5 to 10 minutes. [1]

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- (c) The student used the results from three titrations to calculate a mean titre. Some of her results are shown below.

Titration	1	2	3
Final reading / cm ³	26.90	26.90	
Initial reading / cm ³	0.25	0.15	0.20
Titre / cm ³	26.65	26.75	

Mean titre = 26.73 cm³

Determine the **final reading** for the third titration.

[2]

Final reading = cm³

- (d) Ascorbic acid can decompose upon exposure to air. If this reaction occurred before the titration was completed, state how it might affect the titration results. Explain your answer. [2]

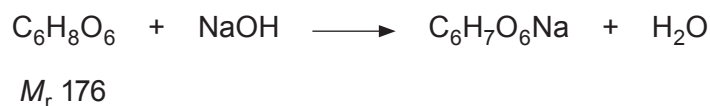
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- (e) The equation for the reaction between ascorbic acid and sodium hydroxide is given below.



The percentage of ascorbic acid is identical in each 500 mg tablet. Calculate the percentage of ascorbic acid in each vitamin C tablet. [3]

Percentage ascorbic acid = %

- (f) Sulfuric acid and hydrochloric acid are strong acids.

- (i) Calculate the pH of a solution of $0.010 \text{ mol dm}^{-3}$ sulfuric acid, H_2SO_4 . [2]

pH =

- (ii) When hydrochloric acid is heated with MnO_2 it reacts according to the following equation.



Explain why this can be classified as a redox reaction. [2]

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SECTION B

Answer **all** questions in the spaces provided.

9. (a) Consider the elements labelled **A-G**. These are not chemical symbols.

A	$1s^2 2s^2 2p^1$
B	$1s^2 2s^2 2p^3$
C	$1s^2 2s^2 2p^6$
D	$1s^2 2s^2 2p^6 3s^2$
E	$1s^2 2s^2 2p^6 3s^2 3p^1$
F	$1s^2 2s^2 2p^6 3s^2 3p^6$
G	$1s^2 2s^2 2p^6 3s^2 3p^6 4s^1$

- (i) Give the letter (**A-G**) of the element with the largest **first** ionisation energy.

Give **two** reasons for your answer.

[3]

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- (ii) Give the letter (**A-G**) of the element with the largest **last** ionisation energy.

Give a reason for your answer.

[2]

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- (iii) Give the letters (**A-G**) of **all** the elements above that are metals.

[2]

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- (iv) Give the letters (**A-G**) of **all** the elements above that form basic oxides.

[1]

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(b) Magnesium exists as three naturally-occurring stable isotopes. They can be identified using a mass spectrometer.

(i) State how magnesium ions are formed in a mass spectrometer. [1]

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(ii) State how magnesium ions are separated in a mass spectrometer. [1]

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(c) Magnesium-28 is an unstable radioactive isotope that decays by β -emission.

(i) Give the mass number and symbol of the element formed as a product of the radioactive decay of magnesium-28. [1]

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(ii) If it takes 84 hours for the activity of the isotope to decay to $\frac{1}{16}$ th of its original activity, calculate its half-life. [1]

Half-life = hours



- (d) Magnesium is a typical metal. Describe the bonding in magnesium and explain why it is ductile. You may include a diagram as part of your answer. [3]

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- (e) According to the label on the bottle, the concentration of magnesium ions in a sample of Welsh mineral water is 15 mg / litre.

Calculate the concentration of magnesium ions in mol dm^{-3} . [1]

Concentration = mol dm^{-3}



9. (a) (i) Explain the origins of emission spectra. [2]

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- (ii) Give **one** difference in the appearance of absorption and emission spectra. [1]

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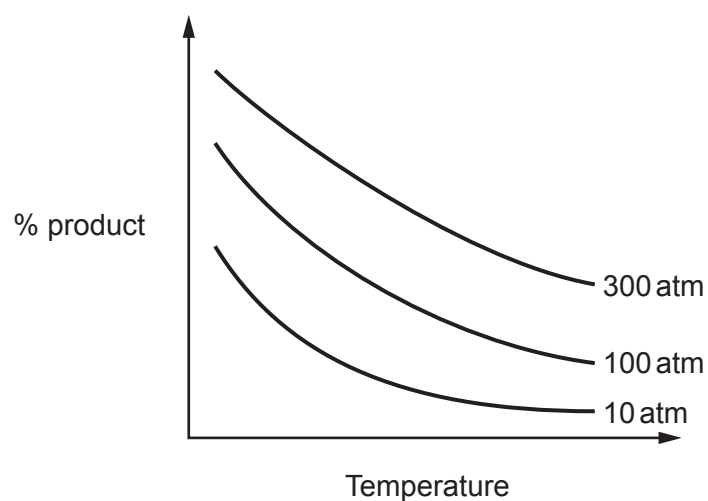
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- (b) A line in the emission spectrum of an element has a wavelength of 95.0 nm. Calculate the frequency of this line in megahertz, MHz. [2]

frequency = MHz



- (c) The diagram below shows how the percentage product varies with temperature and pressure for an equilibrium process.



Use the diagram and Le Chatelier's principle to explain whether:

- (i) the forward reaction is endothermic or exothermic.

[2]

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- (ii) the forward reaction involves an increase or decrease in the number of moles of gas.

[2]

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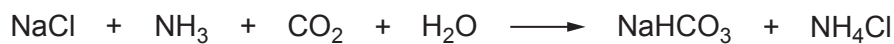
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- (d) The Solvay process is the major industrial process for the manufacture of sodium carbonate. Two of the stages in the process are shown below:



Calculate the maximum mass of sodium carbonate, in kg, which could be obtained from 15.0 tonnes of sodium chloride. Give your answer to an appropriate number of significant figures.

[3]

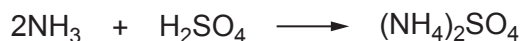
maximum mass = kg

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9. (a) Over 180 million tonnes of ammonia are manufactured each year.

The main use of ammonia is in the production of salts such as ammonium sulfate, which is used as a fertiliser.



Explain why this is an acid-base reaction.

[1]

- (b) Sodium hydroxide reacts with ammonium sulfate to form ammonia, sodium sulfate and water as shown in the equation below.



A 1.86 g sample of ammonium sulfate was neutralised by exactly 26.70 cm³ of a sodium hydroxide solution.

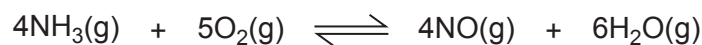
Calculate the concentration, in mol dm⁻³, of the sodium hydroxide solution used.

[3]

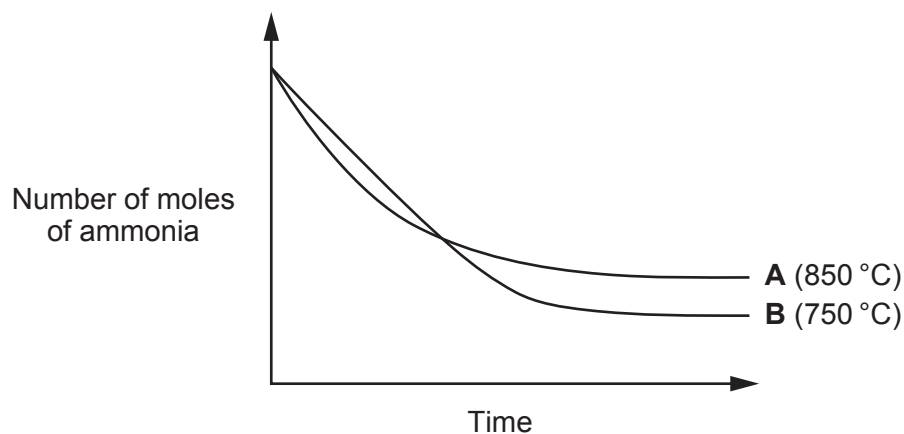
Concentration = mol dm⁻³



- (c) Another use of ammonia is in the production of nitric acid. In the first part of this process ammonia is oxidised in air.



The graph below shows how the number of moles of ammonia present changes as the reaction proceeds until equilibrium is reached.



- (i) State Le Chatelier's principle.

[1]

- (ii) A student looked at curves **A** and **B** and said that the forward reaction is exothermic.

Is he correct? Justify your answer by using Le Chatelier's principle.

[2]

- (iii) On the graph, draw a curve that represents the reaction at 850 °C but with a catalyst added to the reaction mixture.

Label this curve **C**. Explain the shape of the curve.

[2]



- (d) A solution of nitric acid has a concentration of $0.0550 \text{ mol dm}^{-3}$.

Calculate its pH.

[1]

pH =

- (e) A student said that the bonds in an ammonia molecule are not purely covalent.

Explain why she is correct.

[2]

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